ZEARN Math

Georgia State Standards Alignment

For Grades K-8

About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to Georgia’s Mathematics Standards.

We are committed to ensuring that our high-quality instructional materials align to Georgia’s Mathematics Standards to fully support Georgia’s students and teachers. The tables on the following pages, organized by Georgia Mathematics Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
Kindergarten Standards		Lessons
Numerical Reasoning		
K.NR.1.1	Count up to 20 objects in a variety of structured arrangements and up to 10 objects in a scattered arrangement.	Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic E
K.NR.1.2	When counting objects, explain that the last number counted represents the total quantity in a set (cardinality), regardless of the arrangement and order.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 1, Topic G Mission 1, Topic H Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.NR.1.3	Given a number from 1-20, identify the number that is one more or one less.	Mission 1, Topic G Mission 1, Topic H
K.NR.1.4	Identify pennies, nickels, and dimes and know their name and value.	Grade 1, Mission 6, Topic E
K.NR.2.1	Count forward to 100 by tens and ones and backward from 20 by ones.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E
K.NR.2.2	Count forward beginning from any number within 100 and count backward from any number within 20.	Mission 1, Topic G Mission 5, Topic A Mission 5, Topic B

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
Kindergarten Standards		**Lessons**
		Mission 5, Topic D Mission 5, Topic E
K.NR.3.1	Describe numbers from 11 to 19 by composing (putting together) and decomposing (breaking apart) the numbers into ten ones and some more ones.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.NR.4.1	Identify written numerals 0- 20 and represent a number of objects with a written numeral 0-20 (with 0 representing a count of no objects).	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.NR.4.2	Compare two sets of up to 10 objects and identify whether the number of objects in one group is more or less than the other group, using the words "greater than," "less than," or "the same as".	Mission 3, Topic B Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G Mission 3, Topic H
K.NR.5.1	Compose (put together) and decompose (break apart) numbers up to 10 using objects and drawings.	Mission 1, Topic C Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic G
K.NR.5.2	Represent addition and subtraction within 10 from a given authentic situation using a variety of representations and strategies.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H
K.NR.5.3	Use a variety of strategies to solve addition and subtraction problems within 10.	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic F Mission 4, Topic G Mission 4, Topic H

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
Kindergarten Standards		**Lessons**
K.NR.5.4	Fluently add and subtract within 5 using a variety of strategies to solve practical, mathematical problems.	Mission 4, Topic A
Patterning & Algebraic Reasoning		
K.PAR.6.1	Create, extend, and describe repeating patterns with numbers and shapes, and explain the rationale for the pattern.	Mission 1, Topic H Mission 4, Topic B Mission 4, Topic H Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
K.PAR.6.2	Describe patterns involving the passage of time using words and phrases related to actual events.	*It is recommended that teachers use supplemental materials to fully address this standard.*
Measurement & Data Reasoning		
K.MDR.7.1	Directly compare, describe, and order common objects, using measurable attributes (length, height, width, or weight) and describe the difference.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F Mission 3, Topic H
K.MDR.7.2	Classify and sort up to ten objects into categories by an attribute; count the number of objects in each category and sort the categories by count.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic E Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C
K.MDR.7.3	Ask questions and answer them based on gathered information, observations, and appropriate graphical displays to solve problems relevant to everyday life.	*It is recommended that teachers use supplemental materials to fully address this standard.*
Geometric & Spatial Reasoning		
K.GSR.8.1	Identify, sort, classify, analyze, and compare twodimensional shapes and three-dimensional figures, in different sizes and orientations, using informal language to describe their similarities,	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic A

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
Kindergarten Standards		**Lessons**
	differences, number of sides and vertices, and other attributes.	Mission 6, Topic B
K.GSR.8.2	Describe the relative location of an object using positional words.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic B
K.GSR.8.3	Use basic shapes to represent specific shapes found in the environment by creating models and drawings.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic A Mission 6, Topic B
K.GSR.8.4	Use two or more basic shapes to form larger shapes.	Mission 6, Topic B

1st Grade

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
1st Grade Standards		**Lessons**
Numerical Reasoning		
1.NR.1.1	Count within 120, forward and backward, starting at any number. In this range, read and write numerals and represent a number of objects with a written numeral.	Mission 4, Topic A Mission 6, Topic B
1.NR.1.2	Explain that the two digits of a 2-digit number represent the amounts of tens and ones.	Mission 2, Topic D Mission 4, Topic A Mission 4, Topic B Mission 4, Topic F Mission 6, Topic B
1.NR.1.3	Compare and order whole numbers up to 100 using concrete models, drawings, and the symbols >, =, and <.	Mission 4, Topic B Mission 6, Topic B
1.NR.2.1	Use a variety of strategies to solve addition and subtraction problems within 20.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic G Mission 1, Topic H Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 3, Topic C Mission 3, Topic D Mission 4, Topic E Mission 6, Topic A Mission 6, Topic F
1.NR.2.2	Use pictures, drawings, and equations to develop strategies for addition and subtraction within 20 by exploring strings of related problems.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic F Mission 1, Topic I Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
1st Grade Standards		Lessons
		Mission 2, Topic C Mission 2, Topic D
1.NR.2.3	Recognize the inverse relationship between subtraction and addition within 20 and use this inverse relationship to solve authentic problems.	Mission 1, Topic G Mission 1, Topic H Mission 1, Topic I Mission 2, Topic B Mission 2, Topic C
1.NR.2.4	Fluently add and subtract within 10 using a variety of strategies.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic F Mission 1, Topic I Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D
1.NR.2.5	Use the meaning of the equal sign to determine whether equations involving addition and subtraction are true or false.	Mission 1, Topic E Mission 2, Topic B Mission 2, Topic C
1.NR.2.6	Determine the unknown whole number in an addition or subtraction equation relating to three whole numbers.	Mission 1, Topic D Mission 1, Topic H Mission 2, Topic C
1.NR.2.7	Apply properties of operations as strategies to solve addition and subtraction problem situations within 20.	Mission 1, Topic E Mission 1, Topic F Mission 2, Topic A Mission 2, Topic B
Patterning & Algebraic Reasoning		
1.PAR.3.1	Investigate, create, and make predictions about repeating patterns with a core of up to 3 elements resulting from repeating an operation, as a series of shapes, or a number string.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic F Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 4, Topic D Mission 6, Topic B

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
1st Grade Standards		Lessons
1.PAR.3.2	Identify, describe, and create growing, shrinking, and repeating patterns based on the repeated addition or subtraction of 1s, 2s, 5s, and 10s.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic F Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 4, Topic D Mission 6, Topic B
Geometric & Spatial Reasoning		
1.GSR.4.1	Identify common twodimensional shapes and threedimensional figures, sort and classify them by their attributes and build and draw shapes that possess defining attributes.	Mission 5, Topic A
1.GSR.4.2	Compose two-dimensional shapes (rectangles, squares, triangles, half-circles, and quarter-circles) and threedimensional figures (cubes, rectangular prisms, cones, and cylinders) to create a shape formed of two or more common shapes and compose new shapes from the composite shape.	Mission 5, Topic B
1.GSR.4.3	Partition circles and rectangles into two and four equal shares.	Mission 5, Topic C Mission 5, Topic D
Numerical Reasoning		
1.NR.5.1	Use a variety of strategies to solve applicable, mathematical addition and subtraction problems with one- and two-digit whole numbers.	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic F Mission 6, Topi cC Mission 6, Topic D
1.NR.5.2	Given a two-digit number, mentally find 10 more or 10 less than the number, without having to count; explain the reasoning used.	Mission 2, Topic D Mission 4, Topic A Mission 6, Topic B
1.NR.5.3	Add and subtract multiples of 10 within 100.	Mission 4, Topic C Mission 6, Topic C
Measurement & Data		

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
1st Grade Standards		**Lessons**
1.MDR.6.1	Estimate, measure, and record lengths of objects using non-standard units, and compare and order up to three objects using the recorded measurements. Describe the objects compared.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C
1.MDR.6.2	Tell and write time in hours and half-hours using analog and digital clocks, and measure elapsed time to the hour on the hour using a predetermined number line.	Mission 5, Topic D
1.MDR.6.3	Identify the value of quarters and compare the values of pennies, nickels, dimes, and quarters.	Mission 4, Topic A Mission 6, Topic E
1.MDR.6.4	Ask questions and answer them based on gathered information, observations, and appropriate graphical displays to compare and order whole numbers.	Mission 3, Topic D

2nd Grade

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
2nd Grade Standards		Lessons
Numerical Reasoning		
2.NR.1.1	Explain the value of a threedigit number using hundreds, tens, and ones in a variety of ways.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.NR.1.2	Count forward and backward by ones from any number within 1000. Count forward by fives from multiples of 5 within 1000. Count forward and backward by 10s and 100s from any number within 1000. Count forward by 25s from 0.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 6, Topic A Mission 6, Topic B Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
2.NR.1.3	Represent, compare, and order whole numbers to 1000 with an emphasis on place value and equality. Use >, =, and < symbols to record the results of comparisons.	Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
2.NR.2.1	Fluently add and subtract within 20 using a variety of mental, part-whole strategies.	Mission 1, Topic A Mission 1, Topic B
2.NR.2.2	Find 10 more or 10 less than a given three-digit number and find 100 more or 100 less than a given three-digit number.	Mission 3, Topic G Mission 5, Topic A Mission 5, Topic D
2.NR.2.3	Solve problems involving the addition and subtraction of two-digit numbers using partwhole strategies.	Mission 1, Topic B Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic F

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
2nd Grade Standards		**Lessons**
2.NR.2.4	Fluently add and subtract within 100 using strategies based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 1, Topic B Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C
2.NR.3.1	Determine whether a group (up to 20) has an odd or even number of objects. Write an equation to express an even number as a sum of two equal addends.	Mission 6, Topic D
2.NR.3.2	Use addition to find the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns; write an equation to express the total as a sum of equal addends.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C
Patterning & Algebraic Reasoning		
2.PAR.4.1	Identify, describe, and create a numerical pattern resulting from repeating an operation such as addition and subtraction	Mission 1, Topic B Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic D Mission 5, Topic A Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 6, Topic D Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
2.PAR.4.2	Identify, describe, and create growing patterns and shrinking patterns involving addition and subtraction up to 20.	Mission 1, Topic B Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic D Mission 5, Topic A

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2nd Grade Standards		Lessons
		Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 6, Topic D Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
Measurement & Data Reasoning		
2.MDR.5.1	Construct simple measuring instruments using unit models. Compare unit models to rulers.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 7, Topic C Mission 7, Topic D Mission 7, Topic F Mission 8, Topic A
2.MDR.5.2	Estimate and measure the length of an object or distance to the nearest whole unit using appropriate units and standard measuring tools.	Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 7, Topic D
2.MDR.5.3	Measure to determine how much longer one object is than another and express the length difference in terms of a standard-length unit.	Mission 2, Topic C Mission 2, Topic D Mission 7, Topic D
2.MDR.5.4	Ask questions and answer them based on gathered information, observations, and appropriate graphical displays to solve problems relevant to everyday life.	Mission 7, Topic A
2.MDR.5.5	Represent whole-number sums and differences within a standard unit of measurement on a number line diagram.	Mission 7, Topic A Mission 7, Topic E Mission 7, Topic F
2.MDR.6.1	Tell and write time from analog and digital clocks to the nearest five minutes, and estimate and measure elapsed time using a timeline, to the hour or half hour on the hour or half hour.	Mission 8, Topic D
2.MDR.6.2	Find the value of a group of coins and determine combinations of coins that equal a given amount that	Mission 3, Topic D Mission 7, Topic B

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
2nd Grade Standards		Lessons
	is less than one hundred cents, and solve problems involving dollar bills, quarters, dimes, nickels, and pennies, using $ and ¢ symbols appropriately.	
Geometric & Spatial Reasoning		
2.GSR.7.1	Describe, compare and sort 2-D shapes including polygons, triangles, quadrilaterals, pentagons, hexagons, and 3-D shapes including rectangular prisms and cones, given a set of attributes.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C
2.GSR.7.2	Identify at least one line of symmetry in everyday objects to describe each object as a whole.	Mission 6, Topic C
2.GSR.7.3	Partition circles and rectangles into two, three, or four equal shares. Identify and describe equal-sized parts of the whole using fractional names ("halves," "thirds," "fourths", "half of," "third of," "quarter of," etc.).	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
2.GSR.7.4	Recognize that equal shares of identical wholes may be different shapes within the same whole.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D

3rd Grade

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
3rd Grade Standards		Lessons
Numerical Reasoning		
3.NR.1.1	Read and write multi-digit whole numbers up to 10,000 using base-ten numerals and expanded form.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.NR.1.2	Use place value reasoning to compare multi-digit numbers up to 10,000, using >, =, and < symbols to record the results of comparisons.	Grade 4, Mission 1, Topic B
3.NR.1.3	Use place value understanding to round whole numbers up to 1000 to the nearest 10 or 100.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
Patterning & Algebraic Reasoning		
3.PAR.2.1	Fluently add and subtract within 1000 to solve problems.	Mission 2, Topic A Mission 2, Topic D Mission 2, Topic E
3.PAR.2.2	Apply part-whole strategies, properties of operations and place value understanding, to solve problems involving addition and subtraction within 10,000. Represent these problems using equations with a letter standing for the unknown quantity. Justify solutions.	Grade 4, Mission 1, Topic D Grade 4, Mission 1, Topic E Grade 4, Mission 1, Topic F
3.PAR.3.1	Describe, extend, and create numeric patterns related to multiplication. Make predictions related to the patterns.	Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
3.PAR.3.2	Represent single digit multiplication and division facts using a variety of strategies. Explain the relationship between multiplication and division.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
3rd Grade Standards		**Lessons**
		Mission 3, Topic D Mission 3, Topic E
3.PAR.3.3	Apply properties of operations (i.e., commutative property, associative property, distributive property) to multiply and divide within 100.	Mission 1, Topic C Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F
3.PAR.3.4	Use the meaning of the equal sign to determine whether expressions involving addition, subtraction, and multiplication are equivalent.	Grade 1, Mission 1, Topic E Grade 1, Mission 2, Topic B Grade 1, Mission 2, Topic C
3.PAR.3.5	Use place value reasoning and properties of operations to multiply one-digit whole numbers by multiples of 10, in the range 10-90.	Mission 3, Topic F
3.PAR.3.6	Solve practical, relevant problems involving multiplication and division within 100 using part-whole strategies, visual representations, and/or concrete models.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.PAR.3.7	Use multiplication and division to solve problems involving whole numbers to 100. Represent these problems using equations with a letter standing for the unknown quantity. Justify solutions.	Mission 1, Topic F Mission 3, Topic C Mission 3, Topic E Mission 3, Topic F Mission 7, Topic A
Numerical Reasoning		
3.NR.4.1	Describe a unit fraction and explain how multiple copies of a unit fraction form a non-unit fraction. Use	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
3rd Grade Standards		**Lessons**
	parts of a whole, parts of a set, points on a number line, distances on a number line and area models.	
3.NR.4.2	Compare two unit fractions by flexibly using a variety of tools and strategies.	Mission 5, Topic C
3.NR.4.3	Represent fractions, including fractions greater than one, in multiple ways.	Mission 5, Topic D
3.NR.4.4	Recognize and generate simple equivalent fractions.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Measurement & Data Reasoning		
3.MDR.5.1	Ask questions and answer them based on gathered information, observations, and appropriate graphical displays to solve problems relevant to everyday life.	Mission 6, Topic A
3.MDR.5.2	Tell and write time to the nearest minute and estimate time to the nearest fifteen minutes (quarter hour) from the analysis of an analog clock.	Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.MDR.5.3	Solve meaningful problems involving elapsed time, including intervals of time to the hour, half hour, and quarter hour where the times presented are only on the hour, half hour, or quarter hour within a.m. or p.m. only.	Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.MDR.5.4	Use rulers to measure lengths in halves and fourths (quarters) of an inch and a whole inch.	Mission 6, Topic B Mission 7, Topic D
3.MDR.5.5	Estimate and measure liquid volumes, lengths and masses of objects using customary units. Solve problems involving mass, length, and volume given in the same unit, and reason about the relative sizes of measurement units within the customary system.	Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
Geometric & Spatial Reasoning		

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
3rd Grade Standards		**Lessons**
3.GSR.6.1	Identify perpendicular line segments, parallel line segments, and right angles, identify these in polygons, and solve problems involving parallel line segments, perpendicular line segments, and right angles.	Grade 4, Mission 4, Topic D
3.GSR.6.2	Classify, compare, and contrast polygons, with a focus on quadrilaterals, based on properties. Analyze specific 3- dimensional figures to identify and describe quadrilaterals as faces of these figures.	Mission 7, Topic B
3.GSR.6.3	Identify lines of symmetry in polygons.	Grade 4, Mission 4, Topic D
3.GSR.7.1	Investigate area by covering the space of rectangles presented in realistic situations using multiple copies of the same unit, with no gaps or overlaps, and determine the total area (total number of units that covered the space).	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
3.GSR.7.2	Determine the area of rectangles (or shapes composed of rectangles) presented in relevant problems by tiling and counting.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C
3.GSR.7.3	Discover and explain how area can be found by multiplying the dimensions of a rectangle.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
3.GSR.8.1	Determine the perimeter of a polygon and explain that the perimeter represents the distance around a polygon. Solve problems involving perimeters of polygons.	Mission 7, Topic C Mission 7, Topic D Mission 7, Topic E
3.GSR.8.2	Investigate and describe how rectangles with the same perimeter can have different areas or how rectangles with the same area can have different perimeters.	Mission 7, Topic C Mission 7, Topic D Mission 7, Topic E

4th Grade

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		Lessons
Numerical Reasoning		
4.NR.1.1	Read and write multi-digit whole numbers to the hundred-thousands place using base-ten numerals and expanded form.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
4.NR.1.2	Recognize and show that a digit in one place has a value ten times greater than what it represents in the place to its right and extend this understanding to determine the value of a digit when it is shifted to the left or right, based on the relationship between multiplication and division.	Mission 1, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic G
4.NR.1.3	Use place value reasoning to represent, compare, and order multi-digit numbers, using >, =, and < symbols to record the results of comparisons.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
4.NR.1.4	Use place value understanding to round multi-digit whole numbers.	Mission 1, Topic C
4.NR.2.1	Fluently add and subtract multi-digit numbers to solve practical, mathematical problems using place value understanding, properties of operations, and relationships between operations.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
4.NR.2.2	Interpret, model, and solve problems involving multiplicative comparison.	Mission 1, Topic A Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 5, Topic G Mission 7, Topic A Mission 7, Topic B

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		**Lessons**
4.NR.2.3	Solve relevant problems involving multiplication of a number with up to four digits by a 1-digit whole number or involving multiplication of two two-digit numbers using strategies based on place value and the properties of operations. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic H
4.NR.2.4	Solve authentic division problems involving up to 4-digit dividends and 1- digit divisors (including whole number quotients with remainders) using strategies based on place-value understanding, properties of operations, and the relationships between operations.	Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G
4.NR.2.5	Solve multi-step problems using addition, subtraction, multiplication, and division involving whole numbers. Use mental computation and estimation strategies to justify the reasonableness of solutions.	Mission 1, Topic D Mission 1, Topic E Mission 2, Topic F Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 3, Topic H Mission 7, Topic B Mission 7, Topic C
Patterning & Algebraic Reasoning		
4.PAR.3.1	Generate both number and shape patterns that follow a provided rule.	Mission 5, Topic H
4.PAR.3.1	Use input-output rules, tables, and charts to represent and describe patterns, find relationships, and solve problems.	*It is recommended that teachers use supplemental materials to fully address this standard.*
4.PAR.3.1	Find factor pairs in the range 1–100 and find multiples of single-digit numbers up to 100.	Mission 3, Topic F
4.PAR.3.1	Identify composite numbers and prime numbers and explain the relationship with the factor pairs.	Mission 3, Topic F
Numerical Reasoning		

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		**Lessons**
4.NR.4.1	Using concrete materials, drawings, and number lines, demonstrate and explain the relationship between equivalent fractions, including fractions greater than one, and explain the identity property of multiplication as it relates to equivalent fractions. Generate equivalent fractions using these relationships.	Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 6, Topic B
4.NR.4.2	Compare two fractions with the same numerator or the same denominator by reasoning about their size and recognize that comparisons are valid only when the two fractions refer to the same whole.	Grade 3, Topic 5, Mission C
4.NR.4.3	Compare two fractions with different numerators and/or different denominators by flexibly using a variety of tools and strategies and recognize that comparisons are valid only when the two fractions refer to the same whole.	Mission 5, Topic C Mission 5, Topic E
4.NR.4.4	Represent whole numbers and fractions as the sum of unit fractions.	Mission 5, Topic A
4.NR.4.5	Represent a fraction as a sum of fractions with the same denominator in more than one way, recording with an equation.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F Mission 6, Topic D
4.NR.4.6	Add and subtract fractions and mixed numbers with like denominators using a variety of tools.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F Mission 6, Topic D
4.NR.5.1	Demonstrate and explain the concept of equivalent fractions with denominators of 10 and 100, using concrete materials and visual models. Add two fractions with denominators of 10 and 100.	Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 6, Topic B
4.NR.5.2	Represent, read, and write fractions with denominators of 10 or 100 using decimal notation,	Mission 6, Topic A Mission 6, Topic B

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		**Lessons**
	and decimal numbers to the hundredths place as fractions, using concrete materials and drawings.	Mission 6, Topic D Mission 6, Topic E
4.NR.5.3	Compare two decimal numbers to the hundredths place by reasoning about their size. Record the results of comparisons with the symbols >, =, or <, and justify the conclusions.	Mission 6, Topic B Mission 6, Topic C
Measurement & Data Reasoning		
4.MDR.6.1	Use the four operations to solve problems involving elapsed time to the nearest minute, intervals of time, metric measurements of liquid volumes, lengths, distances, and masses of objects, including problems involving fractions with like denominators, and also problems that require expressing measurements given in a larger unit in terms of a smaller unit, and expressing a smaller unit in terms of a larger unit based on the idea of equivalence.	Mission 2, Topic A Mission 2, Topic B Mission 5, Topic D Mission 5, Topic F Mission 5, Topic G Mission 6, Topic C Mission 6, Topic E Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C
4.MDR.6.2	Ask questions and answer them based on gathered information, observations, and appropriate graphical displays to solve problems relevant to everyday life.	*It is recommended that teachers use supplemental materials to fully address this standard.*
4.MDR.6.3	Create dot plots to display a distribution of numerical (quantitative) measurement data.	Mission 5, Topic E Mission 5, Topic G
Geometric & Spatial Reasoning		
4.GSR.7.1	Recognize angles as geometric shapes formed when two rays share a common endpoint. Draw right, acute, and obtuse angles based on the relationship of the angle measure to 90 degrees.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic D
4.GSR.7.2	Measure angles in reference to a circle with the center at the common endpoint of two rays. Determine an angle's measure in relation to the 360 degrees in a circle through division or as a missing factor problem.	Mission 4, Topic B
4.GSR.8.1	Explore, investigate, and draw points, lines, line segments, rays, angles (right, acute, obtuse),	Mission 4, Topic A Mission 4, Topic D

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		**Lessons**
	perpendicular lines, parallel lines, and lines of symmetry. Identify these in twodimensional figures.	
4.GSR.8.2	Classify, compare, and contrast polygons based on lines of symmetry, the presence or absence of parallel or perpendicular line segments, or the presence or absence of angles of a specified size and based on side lengths.	Mission 4, Topic D
4.GSR.8.3	Solve problems involving area and perimeter of composite rectangles involving whole numbers with known side lengths.	Mission 3, Topic A Mission 3, Topic H

5th Grade

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
5th Grade Standards		Lessons
Numerical Reasoning		
5.NR.1.1	Explain that in a multi-digit number, a digit in one place represents 10 times as much as it represents in the place to its right and 1/10 of what it represents in the place to its left.	Mission 1, Topic A Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
5.NR.1.2	Explain patterns in the placement of digits when multiplied or divided by a power of 10. Use whole-number exponents to denote powers of 10, up to 10^3 .	Mission 1, Topic A Mission 1, Topic D Mission 1, Topic E Mission 2, Topic A Mission 2, Topic D Mission 2, Topic E Mission 2, Topic G
5.NR.2.1	Fluently multiply multi-digit (up to 3- digit by 2-digit) whole numbers to solve authentic problems.	Mission 2, Topic B Mission 2, Topic D
5.NR.2.2	Fluently divide multi-digit whole numbers (up to 4-digit dividends and 2-digit divisors no greater than 25) to solve practical problems.	Mission 2, Topic E Mission 2, Topic F Mission 2, Topic H
5.NR.3.1	Explain the meaning of a fraction as division of the numerator by the denominator *($a/b = a \div b$)*. Solve problems involving division of whole numbers leading to answers in the form of fractions or mixed numbers.	Mission 3, Topic A Mission 4, Topic B
5.NR.3.2	Compare and order up to three fractions with different numerators and/or different denominators by flexibly using a variety of tools and strategies.	Grade 4, Mission 5TC
5.NR.3.3	Model and solve problems involving addition and subtraction of fractions and mixed numbers with unlike denominators.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 4, Topic D

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
5th Grade Standards		**Lessons**
5.NR.3.4	Model and solve problems involving multiplication of a fraction and a whole number.	Grade 4, Mission 5, Topic A Grade 4, Mission 5, Topic E Grade 4, Mission 5, Topic G
5.NR.3.5	Explain why multiplying a whole number by a fraction greater than one results in a product greater than the whole number, and why multiplying a whole number by a fraction less than one results in a product less than the whole number and multiplying a whole number by a fraction equal to one results in a product equal to the whole number.	Mission 4, Topic F
5.NR.3.6	Model and solve problems involving division of a unit fraction by a whole number and a whole number by a unit fraction.	Mission 4, Topic G
5.NR.4.1	Read and write decimal numbers to the thousandths place using baseten numerals written in standard form and expanded form.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
5.NR.4.2	Represent, compare, and order decimal numbers to the thousandths place based on the meanings of the digits in each place, using >, =, and < symbols to record the results of comparisons.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
5.NR.4.3	Use place value understanding to round decimal numbers to the hundredths place.	Mission 1, Topic C
5.NR.4.4	Solve problems involving addition and subtraction of decimal numbers to the hundredths place using a variety of strategies.	Mission 1, Topic D
5.NR.5.1	Write, interpret, and evaluate simple numerical expressions involving whole numbers with or without grouping symbols to represent actual situations.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H Mission 6, Topic B

Patterning & Algebraic Reasoning

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
5th Grade Standards		**Lessons**
5.PAR.6.1	Generate two numerical patterns using two given rules. Identify apparent relationships between corresponding terms by completing a table.	Mission 6, Topic B Mission 6, Topic D
5.PAR.6.2	Represent problems by plotting ordered pairs and explain coordinate values of points in the first quadrant of the coordinate plane.	Mission 6, Topic C Mission 6, Topic D
Measurement & Data Reasoning		
5.MDR.7.1	Explore realistic problems involving different units of measurement, including distance, mass, weight, volume, and time.	Mission 2, Topic D
5.MDR.7.2	Ask questions and answer them based on gathered information, observations, and appropriate graphical displays to solve problems relevant to everyday life.	*It is recommended that teachers use supplemental materials to fully address this standard.*
5.MDR.7.3	Convert among units within the metric system and then apply these conversions to solve multistep, practical problems.	Mission 1, Topic A Mission 2, Topic D
5.MDR.7.4	Convert among units within relative sizes of measurement units within the customary measurement system.	Mission 2, Topic D
Geometric & Spatial Reasoning		
5.GSR.8.1	Classify, compare, and contrast polygons based on properties.	Mission 5, Topic D
5.GSR.8.2	Determine, through exploration and investigation, that attributes belonging to a category of two dimensional figures also belong to all subcategories of that category.	Mission 5, Topic D
5.GSR.8.3	Investigate volume of right rectangular prisms by packing them with unit cubes without gaps or overlaps. Then, determine the total volume to solve problems.	Mission 5, Topic A Mission 5, Topic B

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
5th Grade Standards		Lessons
5.GSR.8.4	Discover and explain how the volume of a right rectangular prism can be found by multiplying the area of the base times the height to solve authentic, mathematical problems.	Mission 5, Topic B

6th Grade

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
Numerical Reasoning		
6.NR.1.1	Fluently add and subtract any combination of fractions to solve problems.	Grade 5, Mission 3, Topic B Grade 5, Mission 3, Topic C
6.NR.1.2	Multiply and divide any combination of whole numbers, fractions, and mixed numbers using a student-selected strategy. Interpret products and quotients of fractions and solve word problems.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E
6.NR.1.3	Perform operations with multi-digit decimal numbers fluently using models and student-selected strategies.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
6.NR.2.1	Describe and interpret the center of the distribution by the equal share value (mean).	Mission 8, Topic B
6.NR.2.2	Summarize categorical and quantitative (numerical) data sets in relation to the context: display the distributions of quantitative (numerical) data in plots on a number line, including dot plots, histograms, and box plots and display the distribution of categorical data using bar graphs.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
6.NR.2.3	Interpret numerical data to answer a statistical investigative question created. Describe the distribution of a quantitative (numerical) variable collected, including its center, variability, and overall shape.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
6.NR.2.4	Design simple experiments and collect data. Use data gathered from realistic scenarios and simulations to determine quantitative measures of center (median and/or mean) and variability (interquartile range and range). Use these quantities to draw conclusions about the data, compare different numerical data sets, and make predictions.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		**Lessons**
6.NR.2.5	Relate the choice of measures of center and variability to the shape of the data distribution and the context in which the data were gathered.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
6.NR.2.6	Describe the impact that inserting or deleting a data point has on the mean and the median of a data set. Create data displays using a dot plot or box plot to examine this impact.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
6.NR.3.1	Identify and compare integers and explain the meaning of zero based on multiple authentic situations.	Mission 7, Topic A
6.NR.3.2	Order and plot integers on a number line and use distance from zero to discover the connection between integers and their opposites.	Mission 7, Topic A
6.NR.3.3	Recognize and explain that opposite signs of integers indicate locations on opposite sides of zero on the number line; recognize and explain that the opposite of the opposite of a number is the number itself.	Mission 7, Topic A
6.NR.3.4	Write, interpret, and explain statements of order for rational numbers in authentic, mathematical situations. Compare rational numbers, including integers, using equality and inequality symbols.	Mission 7, Topic B
6.NR.3.5	Explain the absolute value of a rational number as its distance from zero on the number line; interpret absolute value as distance for a positive or negative quantity in a relevant situation.	Mission 7, Topic A Mission 7, Topic C
6.NR.3.6	Distinguish comparisons of absolute value from statements about order	Mission 7, Topic A
6.NR.4.1	Explain the concept of a ratio, represent ratios, and use ratio language to describe a relationship between two quantities.	Mission 2, Topic A Mission 2, Topic B
6.NR.4.2	Make tables of equivalent ratios relating quantities with whole-number measurements, find missing values in the tables, and plot the pairs of values on the coordinate plane. Use tables to compare ratios.	Mission 2, Topic D

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		**Lessons**
6.NR.4.3	Solve problems involving proportions using a variety of student-selected strategies.	Mission 2, Topic D Mission 2, Topic E
6.NR.4.4	Describe the concept of rates and unit rate in the context of a ratio relationship.	Mission 2, Topic C Mission 3, Topic A Mission 3, Topic C
6.NR.4.5	Solve unit rate problems including those involving unit pricing and constant speed.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E Mission 2, Topic F Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 6, Topic B Mission 6, Topic D
6.NR.4.6	Calculate a percent of a quantity as a rate per 100 and solve everyday problems given a percent.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 6, Topic B Mission 6, Topic D
6.NR.4.7	Use ratios to convert within measurement systems (customary and metric) to solve authentic problems that exist in everyday life.	Mission 3, Topic B Mission 3, Topic C
Geometric & Spatial Reasoning		
6.GSR.5.1	Explore area as a measurable attribute of triangles, quadrilaterals, and other polygons conceptually by composing or decomposing into rectangles, triangles, and other shapes. Find the area of these geometric figures to solve problems.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic G Mission 4, Topic D
6.GSR.5.2	Given the net of three-dimensional figures with rectangular and triangular faces, determine the surface area of these figures.	Mission 1, Topic E Mission 1, Topic F Mission 1, Topic G

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		**Lessons**
6.GSR.5.3	Calculate the volume of right rectangular prisms with fractional edge lengths by applying the formula, V = (area of base) x (height).	Mission 7, Topic C
Patterning & Algebraic Reasoning		
6.PAR.6.1	Write and evaluate numerical expressions involving rational bases and whole-number exponents.	Mission 1, Topic F Mission 6, Topic C
6.PAR.6.2	Determine greatest common factors and least common multiples using a variety of strategies to make sense of applicable problems.	Mission 7, Topic D
6.PAR.6.3	Write and read expressions that represent operations with numbers and variables in realistic situations.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic F Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.PAR.6.4	Evaluate expressions when given values for the variables, including expressions that arise in everyday situations.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic F Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.PAR.6.5	Apply the properties of operations to identify and generate equivalent expressions.	Mission 6, Topic B
6.PAR.7.1	Solve one-step equations and inequalities involving variables when values for the variables are given. Determine whether an equation and inequality involving a variable is true or false for a given value of the variable.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.PAR.7.2	Write one-step equations and inequalities to represent and solve problems; explain that a variable can represent an unknown number or any number in a specified set.	Mission 6, Topic A Mission 6, Topic B Mission 7, Topic B

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
6.PAR.7.3	Solve problems by writing and solving equations of the form $x + p = q$, $px = q$ and $x/p = q$ for cases in which p, q and x are all nonnegative rational numbers.	Mission 6, Topic A Mission 6, Topic B
6.PAR.7.4	Recognize and generate inequalities of the form $x > c$, $x > c$, $x < c$, or $x < c$ to explain situations that have infinitely many solutions; represent solutions of such inequalities on a number line.	Mission 7, Topic B
6.PAR.8.1	Locate and position rational numbers on a horizontal or vertical number line; find and position pairs of integers and other rational numbers on a coordinate plane.	Mission 7, Topic A Mission 7, Topic C
6.PAR.8.2	Show and explain that signs of numbers in ordered pairs indicate locations in quadrants of the coordinate plane and determine how two ordered pairs may differ based only on the signs.	Mission 7, Topic C
6.PAR.8.3	Solve problems by graphing points in all four quadrants of the coordinate plane. Include use of coordinates and absolute value to find distances between points with the same xcoordinate or the same y-coordinate.	Mission 7, Topic C
6.PAR.8.4	Draw polygons in the coordinate plane given coordinates for the vertices; use coordinates to find the length of a side joining points with the same x-coordinate or the same ycoordinate.	Mission 7, Topic C

7th Grade

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
Numerical Reasoning		
7.NR.1.1	Show that a number and its opposite have a sum of 0 (are additive inverses). Describe situations in which opposite quantities combine to make 0.	Mission 5, Topic B
7.NR.1.2	Show and explain p + q as the number located a distance |q| from p, in the positive or negative direction, depending on whether q is positive or negative. Interpret sums of rational numbers by describing applicable situations.	Mission 5, Topic A Mission 5, Topic B
7.NR.1.3	Represent addition and subtraction with rational numbers on a horizontal or a vertical number line diagram to solve authentic problems.	Mission 5, Topic A Mission 5, Topic B
7.NR.1.4	Show and explain subtraction of rational numbers as adding the additive inverse, p – q = p + (–q). Show that the distance between two rational numbers on the number line is the absolute value of their difference and apply this principle in contextual situations.	Mission 5, Topic A Mission 5, Topic B
7.NR.1.5	Apply properties of operations, including part-whole reasoning, as strategies to add and subtract rational numbers.	Mission 5, Topic B
7.NR.1.6	Make sense of multiplication of rational numbers using realistic applications.	Mission 5, Topic C
7.NR.1.7	Show and explain that integers can be divided, assuming the divisor is not zero, and every quotient of integers is a rational number.	Mission 5, Topic C
7.NR.1.8	Represent the multiplication and division of integers using a variety of strategies and interpret products and quotients of rational numbers by describing them based on the relevant situation.	Mission 5, Topic C

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
7th Grade Standards		**Lessons**
7.NR.1.9	Apply properties of operations as strategies to solve multiplication and division problems involving rational numbers represented in an applicable scenario.	Mission 5, Topic C
7.NR.1.10	Convert rational numbers between forms to include fractions, decimal numbers and percentages, using understanding of the part divided by the whole. Know that the decimal form of a rational number terminates in 0s or eventually repeats.	Mission 4, Topic A Mission 5, Topic A Mission 5, Topic C Mission 8, Topic D
7.NR.1.11	Solve multi-step, contextual problems involving rational numbers, converting between forms as appropriate, and assessing the reasonableness of answers using mental computation and estimation strategies.	Mission 3, Topic C Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F Mission 6, Topic A Mission 6, Topic B
Patterning & Algebraic Reasoning		
7.PAR.2.1	Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients.	Mission 6, Topic D
7.PAR.2.2	Rewrite an expression in different forms from a contextual problem to clarify the problem and show how the quantities in it are related.	Mission 6, Topic B
7.PAR.3.1	Construct algebraic equations to solve practical problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p, q, and r are specific rational numbers. Interpret the solution based on the situation.	Mission 5, Topic E Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C
7.PAR.3.2	Construct algebraic inequalities to solve problems, leading to inequalities of the form $px + q > r$, $px + q < r$, $px + q \leq r$, or $px + q \geq r$, where p, q, and r are specific rational numbers. Graph and interpret the solution based on the realistic situation that the inequalities represent.	Mission 6, Topic C

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
7th Grade Standards		**Lessons**
7.PAR.4.1	Compute unit rates associated with ratios of fractions, including ratios of lengths, areas and other quantities measured in like or different units presented in realistic problems.	Mission 2, Topic C Mission 4, Topic A
7.PAR.4.2	Determine the unit rate (constant of proportionality) in tables, graphs (1, r), equations, diagrams, and verbal descriptions of proportional relationships to solve realistic problems.	Mission 2, Topic A Mission 2, Topic B
7.PAR.4.3	Determine whether two quantities presented in authentic problems are in a proportional relationship.	Mission 2, Topic A Mission 2, Topic D
7.PAR.4.4	Identify, represent, and use proportional relationships.	Mission 2, Topic B Mission 3, Topic A
7.PAR.4.5	Use context to explain what a point (x, y) on the graph of a proportional relationship means in terms of the situation, with special attention to the points (0, 0) and (1, *r*) where *r* is the unit rate.	Mission 2, Topic D
7.PAR.4.6	Solve everyday problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 2, Topic A Mission 3, Topic B Mission 3, Topic C
7.PAR.4.7	Use similar triangles to explain why the slope, *m*, is the same between any two distinct points on a nonvertical line in the coordinate plane.	Grade 8, Mission 2, Topic C Grade 8, Mission 3, Topic B Grade 8, Mission 3, Topic C Grade 8, Mission 3, Topic E
7.PAR.4.8	Graph proportional relationships, interpreting the unit rate as the slope of the graph. Compare two different proportional relationships represented in different ways.	Grade 8, Mission 3, Topic A Grade 8, Mission 3, Topic B
7.PAR.4.9	Use proportional relationships to solve multi-step ratio and percent problems presented in applicable situations.	Mission 3, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
7.PAR.4.10	Predict characteristics of a population by examining the characteristics of a representative sample. Recognize the potential limitations and scope of the sample to the population.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
7.PAR.4.11	Analyze sampling methods and conclude that random sampling produces and supports valid inferences.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
7.PAR.4.12	Use data from repeated random samples to evaluate how much a sample mean is expected to vary from a population mean. Simulate multiple samples of the same size.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
Geometric & Spatial Reasoning		
7.GSR.5.1	Measure angles in whole nonstandard units.	Grade 4, Mission 4, Topic B
7.GSR.5.2	Measure angles in whole number degrees using a protractor.	Grade 4, Mission 4, Topic B
7.GSR.5.3	Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve equations for an unknown angle in a figure.	Mission 7, Topic A
7.GSR.5.4	Explore and describe the relationship between pi, radius, diameter, circumference, and area of a circle to derive the formulas for the circumference and area of a circle.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C
7.GSR.5.5	Given the formula for the area and circumference of a circle, solve problems that exist in everyday life.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C
7.GSR.5.6	Solve realistic problems involving surface area of right prisms and cylinders.	Mission 7, Topic C
7.GSR.5.7	Describe the two-dimensional figures (cross sections) that result from slicing three-dimensional figures, as in the plane sections of right rectangular prisms, right rectangular pyramids, cones, cylinders, and spheres.	Mission 7, Topic C

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
7.GSR.5.8	Explore volume as a measurable attribute of cylinders and right prisms. Find the volume of these geometric figures using concrete problems.	Mission 7, Topic C Grade 8, Mission 5, Topic D
Probability Reasoning		
7.PR.6.1	Represent the probability of a chance event as a number between 0 and 1 that expresses the likelihood of the event occurring. Describe that a probability near 0 indicates an unlikely event, a probability around 1/2 indicates an event that is neither unlikely nor likely, and a probability near 1 indicates a likely event.	Mission 8, Topic A
7.PR.6.2	Approximate the probability of a chance event by collecting data on an event and observing its long-run relative frequency will approach the theoretical probability.	Mission 8, Topic A
7.PR.6.3	Develop a probability model and use it to find probabilities of simple events. Compare experimental and theoretical probabilities of events. If the probabilities are not close, explain possible sources of the discrepancy.	Mission 8, Topic A Mission 8, Topic C Mission 8, Topic E
7.PR.6.4	Develop a uniform probability model by assigning equal probability to all outcomes and use the model to determine probabilities of events.	Mission 8, Topic A Mission 8, Topic E
7.PR.6.5	Develop a probability model (which may not be uniform) by observing frequencies in data generated from a chance process.	Mission 8, Topic C
7.PR.6.6	Use appropriate graphical displays and numerical summaries from data distributions with categorical or quantitative (numerical) variables as probability models to draw informal inferences about two samples or populations.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E

8th Grade

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
8th Grade Standards		Lessons
Numerical Reasoning		
8.NR.1.1	Distinguish between rational and irrational numbers using decimal expansion. Convert a decimal expansion which repeats eventually into a rational number.	Mission 8, Topic E
8.NR.1.2	Approximate irrational numbers to compare the size of irrational numbers, locate them approximately on a number line, and estimate the value of expressions.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
8.NR.2.1	Apply the properties of integer exponents to generate equivalent numerical expressions.	Mission 7, Topic B Mission 7, Topic C
8.NR.2.2	Use square root and cube root symbols to represent solutions to equations. Recognize that x2 = p (where p is a positive rational number and $\lvert x\rvert \leq 25$) has two solutions and x3 = p (where p is a negative or positive rational number and $\lvert x\rvert \leq 10$) has one solution. Evaluate square roots of perfect squares ≤ 625 and cube roots of perfect cubes ≥ -1000 and ≤ 1000.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
8.NR.2.3	Use numbers expressed in scientific notation to estimate very large or very small quantities, and to express how many times as much one is than the other.	Mission 7, Topic C Mission 7, Topic D
8.NR.2.4	Add, subtract, multiply and divide numbers expressed in scientific notation, including problems where both decimal and scientific notation are used. Interpret scientific notation that has been generated by technology (e.g., calculators or online technology tools).	Mission 7, Topic C Mission 7, Topic D
Patterning & Algebraic Reasoning		
8.PAR.3.1	Interpret expressions and parts of an expression, in context, by utilizing formulas or expressions with multiple terms and/or factors.	Grade 6, Mission 6, Topic B Grade 6, Mission 6, Topic C

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
8th Grade Standards		**Lessons**
8.PAR.3.2	Describe and solve linear equations in one variable with one solution ($x = a$), infinitely many solutions ($a = a$), or no solutions ($a = b$). Show which of these possibilities is the case by successively transforming the given equation into simpler forms, until an equivalent equation of the form $x = a$, $a = a$, or $a = b$ results (where a and b are different numbers).	Mission 4, Topic B
8.PAR.3.3	Create and solve linear equations and inequalities in one variable within a relevant application.	Mission 4, Topic B
8.PAR.3.4	Using algebraic properties and the properties of real numbers, justify the steps of a one-solution equation or inequality.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.PAR.3.5	Solve linear equations and inequalities in one variable with coefficients represented by letters and explain the solution based on the contextual, mathematical situation.	Mission 4, Topic B
8.PAR.3.6	Use algebraic reasoning to fluently manipulate linear and literal equations expressed in various forms to solve relevant, mathematical problems.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.PAR.4.1	Use the equation $y = mx$ (proportional) for a line through the origin to derive the equation $y = mx + b$ (non-proportional) for a line intersecting the vertical axis at b.	Mission 2, Topic C Mission 3, Topic B Mission 3, Topic C Mission 3, Topic E
8.PAR.4.2	Show and explain that the graph of an equation representing an applicable situation in two variables is the set of all its solutions plotted in the coordinate plane.	Mission 3, Topic D
Functional & Graphical Reasoning		
8.FGR.5.1	Show and explain that a function is a rule that assigns to each input exactly one output.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic E
8.FGR.5.2	Within realistic situations, identify and describe examples of functions that are linear or nonlinear.	Mission 5, Topic B Mission 5, Topic C

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8th Grade Standards		**Lessons**
	Sketch a graph that exhibits the qualitative features of a function that has been described verbally.	
8.FGR.5.3	Relate the domain of a linear function to its graph and where applicable to the quantitative relationship it describes.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.FGR.5.4	Compare properties (rate of change and initial value) of two functions used to model an authentic situation each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions).	Mission 5, Topic B Mission 5, Topic C
8.FGR.5.5	Write and explain the equations $y = mx + b$ (slope-intercept form), $Ax + By = C$ (standard form), and $(y - y1) = m(x\text{-}x1)$ (point-slope form) as defining a linear function whose graph is a straight line to reveal and explain different properties of the function.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic E
8.FGR.5.6	Write a linear function defined by an expression in different but equivalent forms to reveal and explain different properties of the function.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.FGR.5.7	Construct a function to model a linear relationship between two quantities. Determine the rate of change and initial value of the function from a description of a relationship or from two (x,y) values, including reading these from a table or from a graph.	Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
8.FGR.5.8	Explain the meaning of the rate of change and initial value of a linear function in terms of the situation it models, and in terms of its graph or a table of values.	Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
8.FGR.5.9	Graph and analyze linear functions expressed in various algebraic forms and show key characteristics of the graph to describe applicable situations.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.FGR.6.1	Show that straight lines are widely used to model relationships between two quantitative variables. For scatter plots that suggest a linear association, visually fit a straight line, and informally assess the model fit by judging the closeness of the data points to the line of best fit.	Mission 6, Topic B

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8th Grade Standards		**Lessons**
8.FGR.6.2	Use the equation of a linear model to solve problems in the context of bivariate measurement data, interpreting the slope and intercepts.	Mission 6, Topic B
8.FGR.6.3	Explain the meaning of the predicted slope (rate of change) and the predicted intercept (constant term) of a linear model in the context of the data.	Mission 6, Topic B
8.FGR.6.4	Use appropriate graphical displays from data distributions involving lines of best fit to draw informal inferences and answer the statistical investigative question posed in an unbiased statistical study.	Mission 6, Topic B
8.FGR.7.1	Interpret and solve relevant mathematical problems leading to two linear equations in two variables.	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
8.FGR.7.2	Show and explain that solutions to a system of two linear equations in two variables correspond to points of intersection of their graphs, because the points of intersection satisfy both equations simultaneously.	Mission 3, Topic D Mission 3, Topic E Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
8.FGR.7.3	Approximate solutions of two linear equations in two variables by graphing the equations and solving simple cases by inspection.	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
8.FGR.7.4	Analyze and solve systems of two linear equations in two variables algebraically to find exact solutions.	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
8.FGR.7.5	Create and compare the equations of two lines that are either parallel to each other, perpendicular to each other, or neither parallel nor perpendicular.	*It is recommended that teachers use supplemental materials to fully address this standard.*
Geometric & Spatial Reasoning		
8.GSR.8.1	Explain a proof of the Pythagorean Theorem and its converse using visual models.	Mission 8, Topic C

GEORGIA'S MATHEMATICS STANDARDS		ZEARN MATH
8th Grade Standards		**Lessons**
8.GSR.8.2	Apply the Pythagorean Theorem to determine unknown side lengths in right triangles within authentic, mathematical problems in two and three dimensions.	Mission 8, Topic C
8.GSR.8.3	Apply the Pythagorean Theorem to find the distance between two points in a coordinate system in practical, mathematical problems.	Mission 8, Topic C
8.GSR.8.4	Apply the formulas for the volume of cones, cylinders, and spheres and use them to solve in relevant problems.	Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F